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Large Language Models, Small Labor Market Effects

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Abstract

We examine the early labor market impacts of AI chatbots by linking large-scale, representative adoption surveys to administrative labor records in Denmark. Using difference-in-differences, we estimate precise null effects on earnings and recorded hours at both the worker and workplace levels, ruling out effects larger than 2% two years after. These null results hold for intensive users, early adopters, workplaces with substantial investments, workers reporting large gains, flexible-pay occupations, and early-career jobs. Adoption is linked to occupational switching and task restructuring, but without net changes in hours or earnings. Our findings challenge narratives of imminent disruption from Generative AI.

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to labor market outcomes.

To examine these outcomes, we link our surveys to administrative records on monthly earnings, hours worked, and occupations through December 2024—two years after the launch of ChatGPT. Using a difference-in-differences framework, we compare adopters and non-adopters before and after the arrival of AI chatbots, estimating impacts at both the worker and workplace levels.

We begin by analyzing workers who use AI chatbots. If these tools enhance individual productivity, a natural question is whether such gains translate into higher earnings. Our first key finding is that AI chatbots have had minimal overall impact on workers who use them. Difference-in-differences estimates for earnings, recorded hours, and wages are all precisely estimated zeros, with confidence intervals ruling out average effects larger than 2%. These null results hold even for workers who use the tools daily, report substantial productivity gains, perform new AI-related tasks, and work in firms that encourage and invest in chatbots. Moreover, we find no significant effects in any of our 11 occupations, including those with flexible, decentralized wage-setting.

While chatbots have not led to earnings gains for users, we find a clear link to occupational mobility: adopters are more likely to have transitioned into new occupations after the arrival of AI chatbots. This aligns with experimental evidence that chatbots are especially beneficial for workers with less prior expertise (Brynjolfsson, Li and Raymond, 2025; Noy and Zhang, 2023), and supports the view that Generative AI helps workers extend into new occupations (Autor, 2024). However, the effects are modest—amounting to 4% of full-time equivalent employment—and do not result in net gains in hours or earnings.

Next, we zoom out to the workplace level and examine whether employers that encourage and invest in AI chatbots have experienced differential changes in labor demand. If chatbots substitute for labor, they could lead to slower hiring or trigger layoffs; if they enhance productivity, firms might expand hiring. Yet we find no such differences: workplaces that encourage chatbot use show no differential changes in employment or